Kexin Wei

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AI/ML engineer specializing in medical image segmentation and deep learning, with full-stack skills to take research models from prototype to validated, production-ready pipelines.

Experience

Research Engineer — Mar 2025 — Present

Imperial Global Singapore. — *Singapore*

- **AI Agents**: Building an agentic (Claude tool-use, multi-step planning) penetration-testing system for medical cyber-physical systems.
- **Simulation**: Translated the MATLAB controller to Python and tuned a real-time closed-loop glucose-insulin patient simulator.
- **Full-Stack**: Built a Flask + Vue.js application for real-time simulation and cyberattack visualization.
- **DevOps**: Dockerized and deployed the simulation stack with multi-container orchestration.

Software Team Lead — Mar 2024 — Jan 2025

Creative Medtech Solutions Pte.Ltd. — *Singapore*

- **AI**: Implemented AI-based medical image segmentation, 4D DICOM processing, and US/MRI image registration algorithm.
- **CyberSecurity**: Enhanced security through encryption protocols, SQL injection protection, and streamlined Qt based GUI.
- **Lead**: Led a team of 6 engineers: Git workflow, Scrum adoption, GoogleTest integration, and code-review standards.

Research and Development Engineer — Jun 2022 — Mar 2024

Creative Medtech Solutions Pte.Ltd. — *Singapore*

- **AI**: Developed and validated an AI-based 3D medical image segmentation pipeline for prostate biopsy guidance.
- **Image**: Prototyped kidney segmentation solutions using SAM, MedSAM, and OpenCV for potential implementation.
- **Robot**: Designed respiratory motion tracking algorithm integrating an A*STAR kidney segmentation CNN, verified in vivo across pig and dog trials.
- **Patent**: Filed 3 patents in medical algorithm design: HIFU calibration, respiratory tracking, and time synchronization.

Education

M.Eng in Medical Robotics and AI — Aug 2020 — May 2022

National University of Singapore, Singapore — *GPA: 4.63/5.00*

- Thesis: Deep Reinforcement Learning (DRL) for Robot-assisted Surgical Training. Code.
- TA for ME2143E Motor Characteristics and ME5405 Machine Vision.

B.Eng in Mechanical Engineering — Sep 2014 — Jul 2019

Tongji University, China — *GPA: 4.61/5.00*

Selected Publications

- **K. Wei** et al., "Glucoprint: CGM-Based Re-identification Across Six Type 1 Diabetes Cohorts" (*first author, in prep.*); co-author on *G-FM* (flow-matching time-series generation) and *HealthLoopQA* (clinical QA benchmark), both *submitted 2026*.

Awards & Honors

AI	Hackathon "Runner Up" Prize at bio LLMs hosted by 4Catalyzer (2023)
Mathematics Modeling	2nd Prize in CUMCM (2018); Honorable Mention in MCM/ICM (2018)
Scholarship	3rd Class Award (2018); 1st Class Award (2016); 3rd Class Award (2015)

Skills

AI/ML	PyTorch, TensorFlow, **Deep Learning** (CV, CNN, DRL), **LLM** (Claude, OpenAI, Gemini; LangChain/LangGraph)
Programming	Python, C++, PyTorch, SQL, Docker, Linux, Git, Flask, Vue.js, OpenCV, VTK, MATLAB
CAD/CAE	Solidworks, AutoCAD
Language	**English** (C1), **German** (C1), **Mandarin** (native)